Qualifying Exam in Microeconomics
Arizona State University
June 2023

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. Consider a two-period game with three players, an Owner of a firm, O, and two potential Managers, M_1 and M_2. The firm is an unusual one in that its profit at each date is equal to a positive constant, which I normalize to 1. But to realize this profit at a date, the owner must hire a manager to run the firm at that date.

   At each date, the owner first chooses one of the managers to run the firm. The manager chosen at date t then decides what amount of that date's profit to expropriate. If manager i is chosen at date t and expropriates $x_{it} \in [0, 1]$, then manager i gets x_{it} dollars, the owner gets $1 - x_{it}$ dollars and manager $j \neq i$ gets 0 dollars at date t.

   Before choosing date-2 actions, each player observes the date-1 history, namely, which manager i is chosen, and the amount of profit the chosen manager i expropriates.

   A player's preferences over outcomes of the game are represented by the player's discounted monetary payoff; the common discount factor is $\delta \in (0, 1)$. In what follows confine your attention to pure strategies.

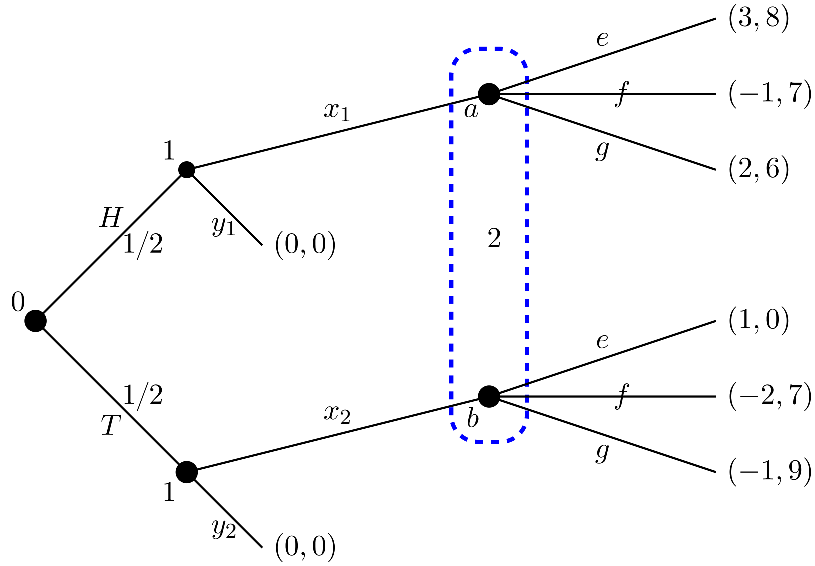
   (a) Describe the strategy sets for the owner and for a manager in this game. Give an example of a strategy for the owner and for a manager.

   (b) What is the highest discounted monetary payoff that the owner can get in a subgame-perfect Nash equilibrium (SPNE)? Identify a subgame-perfect strategy profile that supports this discounted monetary payoff for the owner, and explain clearly why it is subgame perfect and why the owner cannot get a higher discounted monetary payoff.

   (c) What is the lowest discounted monetary payoff that the owner can get in a subgame-perfect Nash equilibrium (SPNE)? Identify a subgame-perfect strategy profile that supports this discounted monetary payoff for the owner, and explain clearly why it is subgame perfect and why the owner cannot get a lower discounted monetary payoff.

   (d) Are there Nash equilibria of this game that are not subgame perfect? If so, identify one and explain why it is a Nash equilibrium and why it is not subgame perfect; if not explain why all Nash equilibria are subgame perfect.

2. Consider the following game in extensive form.



Nature flips a fair coin. Player 1 observes its outcome $z \in \{H, T\}$. Thus player 1 has two information sets; player 2 has a single information set, represented by a dashed box. The top node in player 2's information set is labeled a, the bottom node is labeled b. Payoffs are indicated by number pairs at terminal nodes, the first number corresponds to player 1's payoff. To fix notation, choices at each information set are labeled on the graph.

(a) Let $\alpha \in [0, 1]$ be player 2's beliefs about being in the top node of player 2's information set. At player 2's information set, describe a function that maps every α to player 2's optimal continuation. Briefly link your answer to sequential rationality—note that one sentence suffices to establish the link.

We ask you to *characterize* the set of sequential equilibria for this game. You may directly proceed to part (d) of this problem for full credit. You are also welcome to proceed more piecemeal and start with parts (b)-(c).

(b) Is the strategy profile $[(y_1, y_2), f]$ part of a sequential equilibrium? Justify your answer by providing a proof.

(c) Is the strategy profile (σ_1, σ_2) where

$$\sigma_1(z) = \begin{cases} x_1, & \text{if } z = H \\ (1/7, 6/7) \text{ on } (x_2, y_2), & \text{if } z = T \end{cases}$$

$$\sigma_2 = (2/3, 1/3) \text{ on } (e, f)$$

part of a sequential equilibrium? Justify your answer by providing a proof.

(d) Find *all* sequential equilibria of this game and justify your conclusion.

3. Consider this Robinson Crusoe economy with two goods, one firm, and one consumer. The firm uses good 2 to produce good 1 with a technology given by the production set

$$Y = \{(q, -z) \in \mathbb{R}^2 \mid z \geqslant 0,\ Az \geqslant q\},$$

where A is a positive constant, $z \geqslant 0$ is the quantity of good 2 used as an input, and q is the quantity of good 1 produced as an output. The firm's production function $f : \mathbb{R}_+ \to \mathbb{R}_+$ is therefore $f(z) = Az$.

The consumer's preferences are represented by the function

$$u(x_1, x_2) = 2\sqrt{x_1} + x_2$$

on the consumption set $X = \mathbb{R}^2_+$. Note that the consumption of each good must be *nonnegative.*

The consumer is endowed with $\omega > 0$ units of good 2 and no units of good 1. The consumer owns the firm.

The price of good 2 is normalized to equal 1, and the price of good 1 is $p \geqslant 0$.

(a) Define carefully the notion of a competitive equilibrium for this one-person, private ownership economy.

(b) Derive the supply correspondence of good 1, *carefully* justifying your derivation. Derive the demand correspondence of good 1, again carefully justifying your derivation.

(c) Identify all competitive equilibria for this economy.

(d) To write the obvious, a peculiarity of the Robinson Crusoe economy is that there is only one firm and one consumer. In what follows we do not ask you to prove anything. But if you know of a result or fact which bears on your answer, please mention it. You should set aside the particular functional form assumptions of this problem, and simply address how restrictive it is to assume a single "representative" firm in general.

> As far as determining aggregate profit-maximizing production plans, is there any loss of generality in assuming that there is a single representative firm that is endowed with the aggregate production set? Explain ***in a sentence or two***.

4. A monopolist produces a single good at zero cost. It sells the good to a consumer whose preferences are private information to the consumer. The monopolist offers the consumer a set of contracts, each consisting of a quantity of the good q and a transfer T that the consumer pays the monopolist. The consumer's preferences over contracts are indexed by a real number θ in a finite set $\Theta = \{\theta_1, ...\theta_n\}$ with $\theta_1 < \theta_2 < ... < \theta_n$ with $n \geqslant 2$. A type-θ consumer's preferences over contracts in $\mathbb{R}^2_+$ are represented by the quasilinear function

$$u(q, T, \theta) = f(q, \theta) - T,$$

where $f(\cdot, \theta)$ is strictly increasing on $\mathbb{R}_+$ for every $\theta \in \Theta$.

(a) Prove that u satisfies the *strict single crossing property* in $\big((q, T), \theta\big)$ if and only if f satisfies *strictly increasing differences* in (q, θ)

For the rest of the problem assume that f satisfies strictly increasing differences in (q, θ), and that the equivalence asserted in part (a) is true.

(b) Let C be a nonempty, compact set of contracts in $\mathbb{R}^2_+$ and suppose that (q_i, T_i) maximizes $u(q, T, \theta_i)$ on C for $i = 1, ..., n$. Prove, either directly, or by appealing to an applicable theorem, that $\theta_j > \theta_i$ implies that $(q_j, T_j) \geqslant (q_i, T_i)$.

(c) Specialize to $n = 2$ and suppose that the monopolist believes each type has positive probability. The monopolist chooses $((q_1, T_1), (q_2, T_2)) \geqslant 0$ to maximize expected profit subject to the constraint that, for every $\theta_i \in \Theta$, (q_i, T_i) solves

$$\max u(q, T, \theta_i)$$

$$s.t.\ (q, T) \in \{(0, 0), (q_1, T_1), (q_2, T_2)\}.$$

Let $((q_1^*, T_1^*), (q_2^*, T_2^*))$ be any solution to the monopolist's problem and set $(q_0^*, T_0^*) = (0, 0)$.

Prove that for $i = 1, 2$, a type θ_i consumer must be indifferent between (q_i^*, T_i^*) and (q_{i-1}^*, T_{i-1}^*).